Entropy-Clock Counter-rotating Co-genesis: Baryons and ~1.78 GeV Asymmetric Dark Matter from Pre-inflationary Initial Conditions

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Abstract

We present a mechanism in which the baryon asymmetry and dark matter share a single origin fixed by *pre-inflationary initial conditions* rather than thermal freeze-out. Two counter-rotating hidden condensates carry a $\mathrm{U}(1)_{\mathcal{Q}}$ charge with equal and opposite sign — we postulate $Q_V = -Q_D$ — so that the comoving *difference* charge $a^3(n_V - n_D)$ is exactly conserved and the visible and dark sectors leave a first-order, CP-odd transition at a nucleation temperature $T_n \approx 3.2 \times 10^{12}$ GeV with equal-and-opposite chemical potentials. This *counter-rotating co-genesis* (ECCG) sits in the Affleck–Dine / asymmetric-dark-matter (ADM) class [1–4] with a darkogenesis-style first-order impulse [5, 6] — not a new mechanism *class* — but the architecture is specific and makes a sharp, falsifiable commitment: **dark matter is *asymmetric* and light,** $m_X \approx 1.78$ **GeV**, a GeV-scale direct-detection and collider target, so the model is excluded if dark matter is established as symmetric/thermal or far from the GeV window. Throughout, we are explicit about which statements are calculated here, which are fixed by matching data, and which are postulated, assumed, constructed, or preliminary. Standard-Model sphaleron reprocessing transfers a fixed fraction $f_B = 28/79$ [7] of the $B{-}L$ asymmetry into baryons; we show this textbook value carries over to ECCG because the dark sector is Standard-Model-singlet and chemically decoupled at the electroweak scale ($\Gamma_{\rm mix}/H = 7.5 \times 10^{-4}$), making $f_B = 28/79$ the one genuinely calculated closure of the mechanism — argued (not re-derived operator-by-operator) to carry over unmodified. The shared asymmetry then fixes $m_X = (\Omega_X/\Omega_b)\, m_p\,(28/79) \approx 1.78$ GeV by matching the observed $\Omega_{\rm DM}/\Omega_b$ (matched, not predicted independently). A distinct realization — "Candidate C" — ties $\eta_B \propto T_{\rm RH}^2\,(\Sigma m_\nu)^2$ to neutrino masses; hybridized with ECCG's reheating window it squeezes to **normal ordering,** $\Sigma m_\nu \approx 59$–64 **meV, and** $T_{\rm RH} \approx 3.3 \times 10^{12}$ **GeV** [8]. We are explicit that this range *brushes the normal-ordering floor* ($\Sigma m_\nu \gtrsim 58$ meV): it is a **consistency constraint, not a discriminating prediction** — its confirmation would not single out ECCG, and its inverted-ordering falsification is DESI's own conclusion, independent of this model. It survives in only ~1% of prior impulse-volume, so it constrains the mechanism without testing it against the null. The magnitude of η_B is irreducibly a fit; the microscopic strong-coupling inputs (the first-order transition, the confining vacuum, the wall velocity) are validated pipelines producing preliminary numbers, not confirmed results. We say so throughout, and list the falsifiers.

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I. INTRODUCTION

Two of cosmology's stubborn number problems are the baryon asymmetry $\eta_B \equiv n_B/n_\gamma = 6.1 \times 10^{-10}$ and the dark-matter density $\Omega_{\rm DM}h^2 \approx 0.12$, with the further nagging fact that $\Omega_{\rm DM}/\Omega_b \approx 5$ is an *order-unity* ratio rather than the many-decade gap one would naively expect from unrelated sectors [9, 10]. Asymmetric dark matter (ADM) frameworks [1–3] explain the coincidence by sourcing both sectors from one asymmetry; the price is usually a new mechanism to communicate it. This paper is the matter-genesis face of a larger program — the Unified Structural-Entropy Cosmogenesis (USC) — which posits a single horizon-entropy "clock" $\mathcal{D}_E = (3/2)(1-w)$ joining three physical faces: **dusk** (dark energy without Λ; [Paper III / SEDE cosmology, Zenodo 10.5281/zenodo.21651614]), **dawn** (this paper), and **galactic** (a MOND readout; Paper VIII / APDM galactic, Zenodo 10.5281/zenodo.21652176). The scale sector [Paper VI / scale sector, Zenodo 10.5281/zenodo.21652167] sets the mass scales that this paper's dark-matter horns draw on. The full unification is formalized in the umbrella paper [Paper V / USC framework], and every paper cites the frozen pre-registration falsifier matrix (Zenodo `10.5281/zenodo.21415326`).

For the reader who has not seen those companions: **this paper is self-contained.** The mechanism of §II, the sphaleron transfer of §III, and the central result — the asymmetric dark-matter mass $m_X = 1.782\, f_{\rm sph}(T_{\rm dec}) \approx 1.78$ GeV, fixed by sphaleron timing rather than by washout — are derived here from the stated hidden-sector content and take no numerical input from any companion. The program references above place the construction in its wider setting; the scale-sector paper additionally supplies an *optional* alternative horn (a glueball subcomponent, §VII), which is one branch among those enumerated and not a premise of the main line. Nothing in §§II–V depends on it. All companions are available as open preprints at the DOIs cited, so every reference here resolves to a readable manuscript.

The dawn mechanism, ECCG (Entropy-Clock Counter-rotating Co-Genesis), works as follows. A hidden gauge sector $\mathrm{SU}(3)_H$ supports two condensates carrying opposite $\mathrm{U}(1)_\mathcal{Q}$ charges ($Q_V = -Q_D$). A first-order transition in the early universe gives these condensates a CP-odd impulse; the equal-and-opposite charge structure means the visible and dark sectors leave the transition with equal-and-opposite chemical potentials. The Standard-Model electroweak sphaleron then converts a calculable fraction of the resulting $B-L$ into baryon number, and the leftover dark charge freezes out as asymmetric dark matter whose *mass is fixed by the shared asymmetry* — the ADM co-genesis logic in its cleanest form [11, 12]. The counter-rotating condensate architecture — a charge carried with equal-and-opposite sign in the visible and dark sectors, so that a comoving asymmetry is conserved and the two sectors inherit opposite chemical potentials — is closest in spirit to Affleck-Dine cogenesis [4]; the hidden first-order-transition route from which the impulse is sourced parallels darkogenesis [5] and asymmetric matter from a dark first-order phase transition [6].

We are deliberately austere about what this buys. ECCG is best described as a **matched-EFT existence proof** of GeV-scale asymmetric co-genesis with one sharp handle ($m_X \approx$ 1.78 GeV), not a fully predictive theory: the *sign* of the asymmetry is environmental, the *magnitude* of η_B is a fit, and the microscopic strong-coupling inputs are benchmark-grade. The value of the construction is (i) the single genuinely-calculated closure ($f_B = 28/79$), (ii) the near-floor neutrino *consistency constraint* of the Candidate-C variant, and (iii) the way the dawn face over-determines itself against the rest of the program through the shared clock and the identity $\omega_b \equiv \eta_B$.

A. The connection to the entropy clock — and an important negative result

The wider program is organized around an entropy clock $\Theta_E = \ln(3H/4Gs)$, $\mathcal{D}_E = (3/2)(1-w)$, which reads $\mathcal{D}_E \approx 1$ at the dawn (radiation-like) and $\mathcal{D}_E \to 3$ at the dusk (de Sitter-like). It is tempting — and the early ECCG development attempted this — to have the clock *directly source* the matter asymmetry through a spontaneous-baryogenesis chemical potential $\mu_Q/T \propto H/T$. **That direct-source mechanism is dead** (a calculated, proven negative): the arrow operator that would source it is $\mathbb{Z}_2$-even, so its coefficient $R_0 = 0$ by symmetry, and independent yield estimates fall short by many orders of magnitude. We state this plainly because it is load-bearing for the program's honesty: *the clock does not source matter.* ECCG is a **pre-inflationary initial-condition theory** — the asymmetry is laid down by the condensate impulse at a first-order transition, not pumped by the clock — and the clock's role in the dawn is confined to the timing/thermodynamic backdrop and

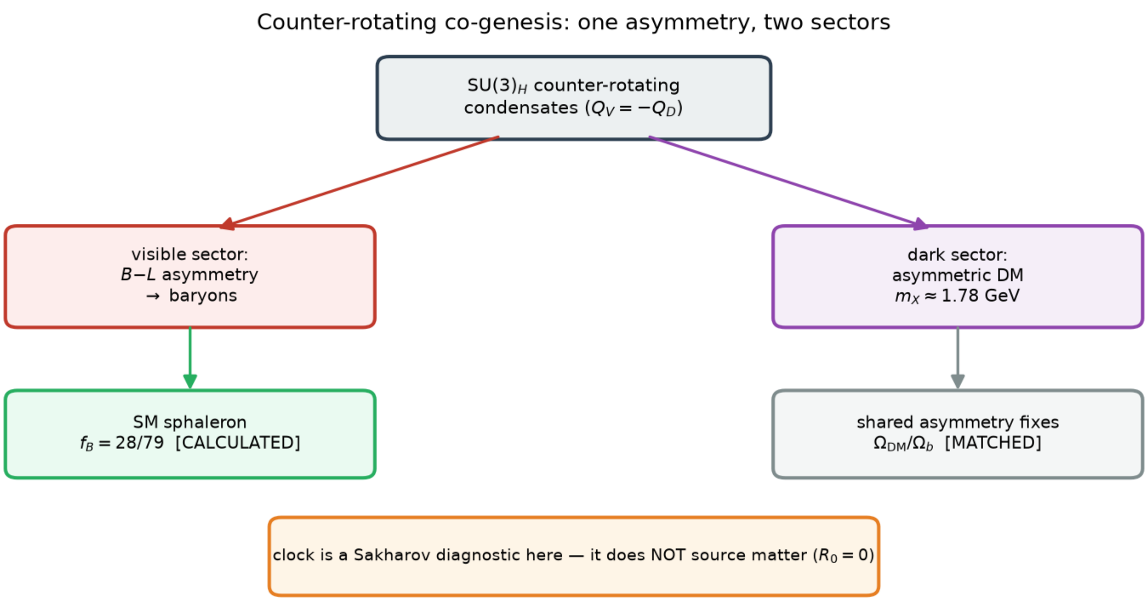

FIG. 1. Counter-rotating co-genesis. SU(3)$_H$ counter-rotating condensates ($Q_V = -Q_D$) seed a $B-L$ asymmetry that the SM sphaleron transfers to baryons with the calculated fraction $f_B = 28/79$, and a shared asymmetry in the dark sector fixing an asymmetric dark-matter density ($m_X \approx 1.78$ GeV, $\Omega_{\rm DM}/\Omega_b$ matched). The entropy clock is a Sakharov diagnostic here — by the $R_0 = 0$ theorem it does not itself source matter.

the over-determination bookkeeping. The $R_0 = 0$ no-go is what forced the program to this honest, narrower position.

II. THE MECHANISM

Figure 1 sets out the mechanism schematically; the rest of this section derives each step.

A. Counter-rotating condensates and the conserved comoving asymmetry

The field content of the ECCG benchmark comprises a hidden SU(3)$_H$ gauge sector, a $\mathbb{Z}_{31}$ flavor structure, a U(1)$_\mathcal{Q}$ under which two scalar condensates Φ_V, Φ_D carry **equal and opposite charge** $Q_V = -Q_D$, which we postulate as the defining structural assumption of the model. The immediate consequence is a conservation law,

$d/dt\,[\,a^3\,(n_V - n_D)\,] = 0$, i.e. $Q_V = -Q_D \Rightarrow$ a conserved comoving *difference* charge (calculated, given the charge assignment).

Two independent ECCG codebases converge on this equal-and-opposite structure, the transfer ratio $R_{\rm tr}$, and m_X (`CROSS_CHECK_MEMO.md`: "AGREE"; $m_X = 1.33$ vs 1.3255 GeV in the older flavored normalization). The counter-rotating structure is what makes this a *co-genesis*: the same impulse that populates the visible $B{-}L$ populates the dark charge, with a fixed relative sign, so the two abundances are locked together. This conserved counter-rotating charge structure — equal-and-opposite $Q_V = -Q_D$ with a conserved comoving difference $a^3(n_V - n_D)$ and equal-and-opposite chemical potentials — is the closest prior art to Affleck-Dine cogenesis [4].

B. The CP-odd impulse and the first-order transition

The asymmetry requires (Sakharov) a departure from equilibrium and CP violation [13]. The departure from equilibrium is a first-order transition; the CP violation enters through the phase of the condensate potential,

$$V_2 = -\Lambda_2^4 \cos(2\psi + \delta_2), \text{ with } \delta_2 = n_2\,\theta_S,\ \delta_3 = n_3\,\theta_S,$$

so that the CP-odd combination $\Delta_{\rm CP} = (3n_2 - 2n_3)\,\theta_S \equiv m \cdot \theta_S$ is set by the **flavon** phase θ_S at scale v_S, not by the hidden sector. Spontaneous CP violation in the minimal $\mathbb{Z}_2$ flavon model makes $\Delta_{\rm CP}$ calculable and generically O(1), peaked at $\pi/2$ (calculated as a distribution, not a single number): the scan gives median $|\Delta_{\rm CP}| = 1.572$ rad, 16–84% $[1.070, 2.067]$, $\sin(\Delta_{\rm CP}/2)$ median 0.708 (`SPONTANEOUS_CP_REPORT`). The symmetric-point value $\Delta_{\rm CP} = \pi/2$ is exact but **not radiatively stable** ($\lambda'' \sim \lambda\lambda'/16\pi^2$ is regenerated), so $\pi/2$ is a central value, not a protected prediction — we quote it as such. Two structural no-go theorems constrain the flavon: a single $\mathrm{U}(1)_{\mathcal{Q}}$-commensurate flavon gives $\Delta_{\rm CP} = 0$ identically, and a single $\mathbb{Z}_N$-invariant harmonic gives only geometric CP-conserving phases $\in \{0, \pi\}$ (calculated and verified numerically).

C. The pre-inflationary domain cure and its reheating threshold (D-1)

A spontaneous-CP mechanism generically produces domains of opposite sign that would wash the net asymmetry to $\eta \approx 0$. ECCG cures this by inflating the flavon phase to uniformity — the domain-selecting quantity is which $\mathbb{Z}_2$ minimum ψ occupies, fixed by $\delta_2 = n_2\theta_S$, and an inflated flavon phase is uniform. This cure survives reheating (calculated, D-1). The worry (critique gap G-M1) was that if $T_{\rm RH}$ exceeds the hidden confinement scale $\Lambda_H = 6.35 \times 10^{12}$ GeV, $\mathrm{SU}(3)_H$ deconfines, Λ_2^4 melts, ψ is released, and reconfinement Kibble-regenerates the domains. It does not fire, for three model-grounded reasons:

1. **The sign lives in the flavon-set phase, not the hidden sector.** The domain-selecting phase is $\delta_2 = n_2\theta_S$, set at v_S, and the model already requires $T_{\rm RH} < v_S$; an inflated flavon phase is uniform.
2. $\mathrm{SU}(3)_H$ **reconfines *real*, adding no phase.** The quantum-modified moduli constraint with positive soft masses selects $\arg\langle\Theta\rangle = 0$, so a melt/reform of the *magnitude* Λ_2 re-pins ψ with the same uniform phase structure.
3. **The released phase cannot random-walk into domains.** Even at the top of the window, the hidden sector is deconfined for only ~0.35 e-folds, and the released ψ fluctuates by $\delta\psi \sim H/(2\pi f_\psi) \approx 2.6 \times 10^{-6}$ rad — six orders below the $\pi/2$ ridge between vacua ($\delta\psi/(\pi/2) = 1.6 \times 10^{-6}$). No flips.

The corrected reheating threshold is therefore $T_{\rm RH} < v_S$**, not** $T_{\rm RH} < \Lambda_H$ (calculated, D-1). The economical single-flavon option (branch B) sits at $v_S = 2.44 \times 10^{13}$ GeV, giving a tight-but-open window $T_{\rm RH} \in [3.2 \times 10^{12}, 2.4 \times 10^{13}]$ GeV (factor 7.7); branch A ($v_S = 4.79 \times 10^{15}$ GeV) is comfortable (factor ~1500) but needs a second gauged flavon to protect strong CP (§VIII, §IX). G-M1 is thereby downgraded from CRITICAL to a satisfied consistency condition — and the cure is not free: it surfaces an independent neutron-EDM falsifier (§IX, branch B).

III. THE SPHALERON TRANSFER $f_B = 28/79$ — THE ONE CALCULATED CLOSURE

Once a $B-L$ asymmetry exists above the electroweak scale, Standard-Model sphalerons reprocess it into a baryon asymmetry with a fixed conversion factor [14]. For the Standard Model with $N_g = 3$ generations and $N_H = 1$ Higgs doublet, the Harvey–Turner equilibrium result [7] is

$f_B = (8N_g + 4N_H)/(22N_g + 13N_H) = 28/79 \approx 0.354$ (a standard textbook Standard-Model result).

The non-trivial ECCG claim is that this *textbook* value carries over unmodified to a theory with an extended dark sector. It does, for two reasons, both computed:

1. **The dark sector is SM-gauge-singlet.** The condensates and X carry no $\mathrm{SU}(2)_L$ isospin, so they are *sphaleron-inert* — the sphaleron acts only on the visible chiral content, and the counting is the pure SM counting.
2. **The dark sector is chemically decoupled at the electroweak scale.** The portal mixing rate is $\Gamma_{\mathrm{mix}}/H = \epsilon^2 \alpha T/H = 7.5 \times 10^{-4} \ll 1$ at $T \approx 130$ GeV (for portal coupling $\epsilon = 4.2 \times 10^{-9}$). So no dark charge leaks back through the sphaleron bath during reprocessing; the visible-baryon factor is the pure SM value.

This is the **one genuinely calculated closure** of the mechanism: an item that moved from "assumed" to "calculated" in the audit (`CLOSURES_assumed_vs_calculated.md`, N4). We flag the honest caveat carried in the corpus: a *strong-portal* or $\mathrm{SU}(2)_L$-charged variant would require recomputation, and the full recomputation of f_B with ECCG's added chiral content at the sphaleron scale has not been done from scratch — it is argued to reduce to the SM value by singlet-ness and decoupling, not re-derived operator-by-operator (`AUDIT_assumed_vs_calculated.md`). Within the benchmark this is safe. We do not inflate it into "the dawn is calculated"; it is one closure, and we present it as the exception, not the rule.

IV. THE DARK-MATTER MASS AND THE CO-GENESIS RELATION

Figure 2 summarises the resulting squeeze on Candidate C, which this section derives.

Given the shared asymmetry and the sphaleron transfer, the dark-matter mass is fixed by requiring the *same* asymmetry to yield the observed $\Omega_{\mathrm{DM}}/\Omega_b$:

$m_X = (\Omega_X/\Omega_b) \cdot m_p \cdot (28/79)/r_{X,BL} \approx 1.78$ GeV at $r_{X,BL} = 1$ (matched to $\Omega_{\mathrm{DM}}/\Omega_b = 5.36$, not predicted) [1].

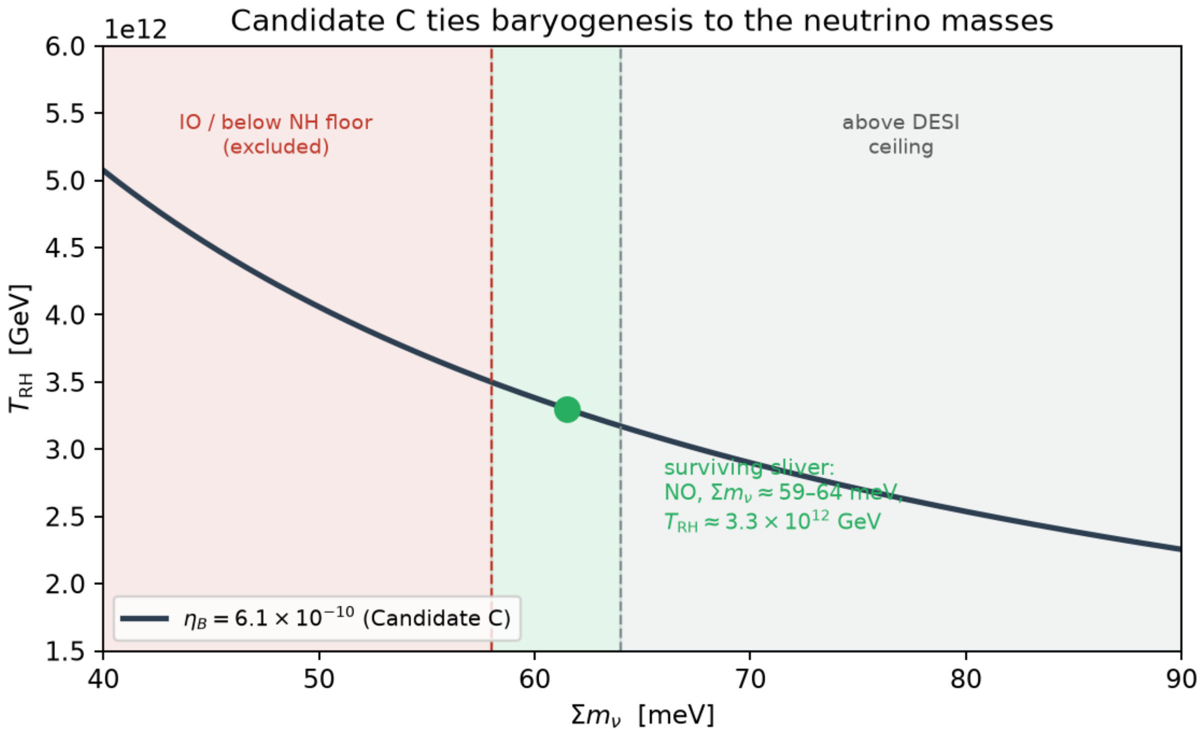

FIG. 2. The Candidate-C squeeze. With $\eta_B \propto T_{\rm RH}^2(\Sigma m_\nu)^2$ fixed to the observed 6.1×10^{-10}, the normal-ordering floor and the DESI ceiling bracket a thin surviving region: normal ordering, $\Sigma m_\nu \approx 59$–64 meV, $T_{\rm RH} \approx 3.3 \times 10^{12}$ GeV. Inverted ordering is excluded; a DESI tightening below the NH floor kills the variant.

This is *matched*, not predicted: $\Omega_{\rm DM}/\Omega_b = 5.36$ is adopted from observation and inverted for the mass. The flavored-charge ratio $r_{X,BL}$ is a genuine model uncertainty — $r_{X,BL} = 1$ gives 1.78 GeV, while the older flavored correction gave ~1.3 GeV (`CONVENTIONS_AND_CORRECTIONS.md`). We adopt $m_X \approx 1.78$ GeV as the benchmark, with the honest note that an earlier momentum-resolved-transport calculation finds m_X set by *sphaleron timing* (not washout), $m_X = 1.782 \cdot f_{\rm sph}(T_{\rm dec})$, landing at 1.78 GeV in ~89% of its viable window (`CONSOLIDATED_THEORY.md`) — a preliminary narrowing (single volume/one-loop, not yet converged) that supports, but does not replace, the matched normalization.

The dark matter is cold, asymmetric, and phenomenologically near-invisible: $\sigma_{\rm SI} \sim 2 \times 10^{-48}$ cm^2, below the neutrino fog. Its most distinctive external signature is *the absence of signatures* — no direct detection, and (in horn (i)) $r \approx 0$ in primordial tensors.

We treat the dark-matter identity as a **three-horn** structure and present all three honestly:

Horn (i) — pure ADM (ECCG-res-iv). The condensate impulse sources both sectors; $\Omega_{\rm DM}/\Omega_b \approx 5$ is *explained* by the shared asymmetry, and $m_X = 1.78$ GeV is sharp. η_B is a fit (§V D). Tensors: $r \leq 2 \times 10^{-11}$. This is the horn in which co-genesis does its signature work — the $\Omega_{\rm DM}/\Omega_b$ coincidence is a consequence, not an input. Its cost is that η_B's magnitude is not derived.

Horn (ii) — Candidate-C hybrid. The clock+Weinberg channel makes η_B calculable (§V) at the cost of discarding the condensate, so m_X becomes a band (0.18–840 GeV) rather than a sharp value, and the $\Omega_{\rm DM}/\Omega_b$ coincidence is one equation in two unknowns. This is the horn that ties η_B to the neutrino sector — a near-floor consistency constraint, not a discriminating prediction (§V).

Horn (iii) — glueball one-sector. The same hidden confining sector that generates the μ scale [Paper VI / scale sector] supplies $\Omega_{\rm DM}$ via 3→2 (SIMP/cannibal) glueball freeze-out, with a *predicted* mass $m_G = 6\Lambda_h = 180$–630 MeV (calculated across the two-loop $\Lambda_h = 28$–105 MeV band). Honest ledger entry (`GLUEBALL_DM_assessment.md`): the SIMP "miracle" here is a **mass** miracle (m_G lands where 3→2 works with a natural coupling $\alpha_{\rm eff} \approx 0.25$–$0.6$), **not an abundance miracle**. Pure glue has no renormalizable portal, so the abundance is set by the hidden-to-visible temperature ratio ξ, and requires an inflaton branching $\mathrm{Br} \approx (0.4\text{–}3) \times 10^{-10}$ selected to ~40% — a dial that *replaces* ADM's explained $\Omega_{\rm DM}/\Omega_b$. This horn uniquely evades the single-source theorem (§X A) because glueball DM is *thermally* sourced, not clock-sourced, so it achieves both a derived η_B (via Candidate C) and a predicted m_G — at the stated cost of the Br dial. Its distinctive signature is velocity-*independent* self-interaction $\sigma/m \approx 0.01$–0.47 cm^2/g, already at the cluster-constraint boundary for $\Lambda_h \approx 30$ MeV.

The environmental sign bit. In all horns the *magnitude* $|\eta_B|$ is predicted but the *sign* is environmental: spontaneous CP violation chooses between $\pm\theta_S^*$ (exactly degenerate CP partners), and inflation makes that choice uniform across the observable universe. So the theory predicts that we live in a matter-dominated (not antimatter-dominated) universe with the observed magnitude, but does not predict *which* — a standard feature of spontaneous-CPV baryogenesis, stated for honesty.

V. CANDIDATE C AND NEUTRINOS (D-2)

A. The channel

Candidate C is the clock-driven variant in which baryogenesis proceeds through the Weinberg operator (whose coefficient *is* the neutrino mass) [15], giving

> $\eta_B \propto T_{\rm RH}^2 \cdot (\Sigma m_i^2)$, more precisely $\eta_B \propto T_{\rm RH}^2\, \Sigma m_\nu^2/2\pi$ (calculated within the matched EFT).

The washout mass is $\bar{m}^2 = \Sigma m_i^2$ (sum of *squares*), while cosmology bounds Σm_i (sum) — different observables. D-2 maps between them via the lightest-mass parametrization for NO and IO, using a real $\Delta L = 2$ washout Boltzmann rather than the weak-washout analytic (which overshoots 13–31% in-window). The calibration: $Y_B(3.17\times 10^{12}, \text{NO-floor}) = 7.98 \times 10^{-11}$, matching the shipped script.

B. The squeeze

Setting $\eta_B = 6.10 \times 10^{-10}$ fixes the line $\eta_B \propto T_{\rm RH}^2\, \Sigma m_i^2$. Intersecting it with the neutrino mass floors, the DESI ceiling, and ECCG's reheating window gives a **razor-thin allowed region** (calculated, D-2):

> $T_{\rm RH} \approx 3.28$–3.31×10^{12} **GeV,** $\Sigma m_\nu \approx 59$–64 **meV, NORMAL ORDERING** (width in $T_{\rm RH}$: factor 1.01).

The NO mass floor (58.6 meV) is matched at $T_{\rm RH} = 3.31 \times 10^{12}$ GeV; the DESI DR2 baseline ceiling (64 meV) [8] is hit at $T_{\rm RH} = 3.28 \times 10^{12}$ GeV. The band is ≤ 6 meV wide and sits exactly at the current DESI cut. The rest of the reheating window ($T_{\rm RH} \gtrsim 3.6 \times 10^{12}$ GeV) needs $\Sigma m_i^2 <$ the NO floor — impossible for any real neutrino spectrum — so it *overproduces* η_B. This is why D-1's widening of the upper window edge (from Λ_H to the portal-thermal 9×10^{12} GeV) does not change the result: the extra room is all overproducing space; the viable sliver at $T_{\rm RH} \approx 3.3 \times 10^{12}$ GeV stands.

C. The two sharp falsifiers, and the ~1% survival

- **Inverted ordering is excluded.** The IO floor ($\Sigma m_\nu = 99$ meV) requires $T_{\rm RH} = 2.4 \times 10^{12}$ GeV (below the reheating window) and lies far above the DESI ceiling [8, 16]. A confirmed IO kills Candidate C.
- **DESI's aggressive bound is already below the NO floor.** The DESI DR2 + DESY5 combination gives $\Sigma m_\nu < 46$ meV [8], *below* the 58.6 meV oscillation floor — the emerging "neutrino mass anomaly." If it holds, the Weinberg operator loses its anchor and Candidate C is falsified together with the mass floor itself.

Honestly, this is a **squeeze, not a parameter-free prediction**. The O(1)-coupling systematic (g_χ, the Weinberg normalization C_W, g_*) is $\approx \times 13$ in $T_{\rm RH}$; the robustness scan finds that *almost every point in that space is already in tension* (raising g_χ overproduces η_B; lowering C_W needs DESI-excluded 100–250 meV neutrinos). Only the canonical point threads the window and a DESI-allowed NO mass — roughly ~1% of prior volume survives. The robust statement is therefore *not* "C predicts $\Sigma m_\nu = 60$ meV" but "C's viable region is squeezed to NO + Σm_ν at its floor + $T_{\rm RH}$ at the bottom of the window, exactly where DESI is cutting." The DESI numbers quoted (baseline 64 meV, aggressive 46 meV) [8] are representative as of early 2026 and should be refreshed against the live posterior before quoting; the *structure* (floor-vs-ceiling squeeze) is robust to the exact ceiling.

D. η_B is matched, not four-times-derived

We are explicit about a meta-pattern the audit flagged (P3). The ECCG corpus contains four "independent confirmations" of η_B — from $\Delta_{\rm CP}$, m_3/H, η_2, and v_w — but each of these is computed through a **common calibrated** `predict_eta_B.py` that already encodes the η_B fit; each report tunes its own knob ($\eta_3 = 1$, m_3/H solved, bracket $= 1$, T_* free) to reproduce the same $\eta_B \approx 6 \times 10^{-10}$. So "four confirmations landing on η_B" is largely **one fit viewed four ways**. The magnitude of η_B (equivalently m_3/H) is irreducibly a fit: an attempt to derive it from maximum-entropy production overshoots by ~10^9 (`USC_m3_derivation.md`). In fairness, the natural (untuned) forward value is not absurd — the spontaneous-CP scan puts the *median* $\eta_B = 1.21 \times 10^{-9}$ (16–84% $[0.87, 1.47] \times 10^{-9}$), within a factor $\simeq 2$ of the observed 6.1×10^{-10}. We state the position of the observed value precisely, because a factor of two is not the same as agreement: recomputing directly from the 40,000-draw scan (`src/spontaneous_cp.py`, `data/spontaneous_cp_scan.csv`), the observed η_B sits at the **6.6th percentile** of the natural distribution — below the 16th percentile, i.e. in the lower tail rather than the bulk. So the honest statement is bounded on both sides: the model's untuned prediction is the right order of magnitude and is *not* fine-tuned to O(1), but it systematically **overshoots** the observed asymmetry by $\simeq 2$, and only $\sim 7\%$ of the natural parameter space lands at or below the measured value. That residual factor is unexplained: it is the size of a dilution or transfer factor, and identifying its origin — or establishing that none exists — is the sharpest open question in this sector. It is *not* a derivation of the magnitude, and m_3/H is still solved to hit η_B. We therefore present $\eta_B = 6.1 \times 10^{-10}$ as **matched, not predicted** (with the rider that the natural value is within $\tilde{\ }\times 2$), and the dawn cross-checks as *one consistency, not four predictions*. What is *not* a fit is the neutrino squeeze of §V B, which follows once the fitted η_B is combined with the *measured* Σm_ν^2 and the reheating window — that is the genuine content of Candidate C.

VI. HONEST STATUS OF THE LOAD-BEARING STRONG-COUPLING INPUTS

The mechanism rests on three strong-coupling inputs that cannot be closed at desk scale. Their honest status is the single most over-claimed thing in prior program-level summaries, and we correct it here. Each is now a fully-specified external project with a decision rule (`EXTERNAL_CALC_SPECS.md`, items S-1/S-2/S-3).

6.1 The first-order transition (S-1). The 4D one-loop and 3D dimensional-reduction legs are robust and give $S_3/T_n = 52.3$, $\beta/H = 347$, $\alpha = 0.027$ at $T_n \approx 3.2\times 10^{12}$ GeV (calculated, within the matched EFT). The *lattice* leg is a **VALIDATED PIPELINE** (agreeing at 1.3σ with one published KKLP point, $\beta_{Hc} = 0.340879(7)$ vs digitized $0.340868(5)$) producing a **PRELIMINARY** ECCG number at *one volume* ($8^2 \times 40$), *one spacing*, with one input digitized from a figure. At the ECCG point the pipeline shows a clean double-peak R^2 histogram (a qualitative first-order signal) and $\beta_{Hc} = 0.344137(9)$, $y3_c = 0.215 \pm 0.003$ (stat) — but with **no $V \to \infty$ or $a \to 0$ extrapolation**, and a ± 0.03 dimensional-reduction systematic that dominates the statistical error. The double-peak character is a robust qualitative result; the *number* is explicitly not a converged critical point. We therefore write "4D one-loop + 3D DR robust; lattice pipeline validated, ECCG point preliminary — continuum / infinite-volume convergence open.” We do **not** write “first-order confirmed by lattice”; that is an overclaim. The S-1 spec (9 volume/spacing combinations, susceptibility scaling $\propto V$ for first-order vs saturation for crossover, ~10^6 core-hours) has an explicit decision rule: **a crossover verdict at any spacing falsifies the mechanism as constructed**, because it removes the out-of-equilibrium leg (§IX, F4).

6.2 The SQCD confining vacuum (S-2). The absolute scales of the matter sector — Λ_H, $\langle\Theta\rangle$, κ_2 — come from a **one-loop qualitative** estimate at strong coupling ($g_H \approx 1.95$). This is a benchmark-level qualitative estimate, not a controlled value (preliminary and qualitative — not yet converged). The quantum-modified moduli constraint with positive soft masses selects the diagonal branch $|M_{ii}| = \Lambda_H^2$, hence $\arg\langle\Theta\rangle = 0$ — a *structural*, assumed input to the D-1 domain cure (§II C). S-2’s decision rule matters downstream: if the honest band on $\langle\Theta\rangle$ moves the derived η_B by more than the reheating-window width, the Candidate-C neutrino squeeze shifts and must be re-quoted; the falsifier currently carries this uncertainty silently.

6.3 The wall velocity (S-3). $v_w = 0.58$ is a **leading-order ballistic** friction estimate (near-sonic deflagration/hybrid; the ± 0.09 is an assumed NLO envelope, not computed) — preliminary and leading-order, not yet converged. At LO the runaway check passes with margin $\times 2.40$ ($\Delta V(T_n) = 0.642\,T^4$ vs max net friction $1.540\,T^4$), and the older benchmark $v_w = 0.30$ is excluded at LO. The paper’s η_B uses $v_w = 0.58$ as if converged; it is not. The S-3 decision rule: $v_w \in [0.3, 0.7]$ keeps η_B stable within ~$\times 2$ (claim survives); $v_w \to 1$ (runaway) drops the sourced asymmetry sharply and forces a re-fit of m_3/H (a fine-tuning flag); $v_w < 0.1$ tightens the DESI Σm_ν squeeze.

VII. THE REVISED BENCHMARK UNDER THE RESOLUTION

For the record, several quantities shifted under the resolution that produced the current benchmark, and un-updated corpus documents may still carry the old values (a bookkeeping caveat, not a physics change): m_X : 1.30 $\to$ 1.78 GeV; f_B : 0.259 $\to$ 28/79 = 0.354; m_3/H : 1.84 $\to$ 1.58; v_w : 0.30 $\to$ 0.58 (estimated). We quote the resolved values throughout.

VIII. THE GAUGED-CP Pin^+ ν mod 16 ANOMALY

Gauging CP (making the spontaneous-CP construction consistent as a gauge theory in 3+1d with Pin^+ structure) carries a $\mathbb{Z}_{16}$-valued Dai–Freed anomaly ($\Omega_5^{\mathrm{Pin}^+} \supset \mathbb{Z}_{16}$; Fidkowski–Kitaev, Witten): each Majorana fermion contributes $\nu = \pm 1$ mod 16, and consistency requires the total $\nu = 0$ mod 16. We **exhibit an explicit satisfying assignment** (an explicit construction, weaker than a derivation): the ~16 CP-relevant Weyl fermions of the R3 realization pair up under the report's own massability structure (four R3 spectator pairs, the condensate-ino conjugate pairs, the $\mathbb{Z}_{31}$ spectators), each pair carrying opposite intrinsic CP phases $(+1, -1)$ and contributing 0, for a total $\nu = 0$ mod 16 (`PIN16_assignment_constructed.md`, `S3_GAUGING_REPORT`). This is *stronger than "satisfiable"* — the per-pair sign flip never changes the sum, so $\nu = 0$ is automatic given CP-commuting mass dressings, and each pair has its own spurion channel so the phases are absorbable one pair at a time. But we say plainly: **this is a construction, not a derivation.** It shows a consistent assignment exists; it does not show one is forced, and a future enlargement of the chiral content re-opens the count.

IX. FALSIFIERS

The dawn face is over-determined by several handles that are *observationally independent* of one another and of the rest of the program. We number the falsifiers; each maps to a pre-registered entry (Zenodo `10.5281/zenodo.21415326`).

- **F1 — DESI Σm_ν.** If DESI DR2/DR3 tightens Σm_ν below the ~59 meV NO floor [8], Candidate C (horns ii, iii) is falsified. This is the union's most imminent make-or-break test; the current baseline (64 meV) already sits at the top of the allowed sliver, and the aggressive combination (46 meV) is already below the floor.
- **F2 — Inverted ordering.** A confirmed IO neutrino spectrum kills Candidate C outright (§V C).

- **F3 — Neutron EDM (branch B).** The D-1 cure forces $T_{\rm RH} < v_S$; the economical branch-B flavon ($v_S = 2.44 \times 10^{13}$ GeV) sits at the nEDM edge, $\bar{\theta} \sim (v_S/M_{\rm Pl})^2 \sin\Delta_{\rm CP} \sim 4 \times 10^{-12} \cdot \mathrm{O}(1)$, predicting a neutron EDM near $\sim 10^{-27}$ $e \cdot cm$ — a DESI/CMB-independent handle. (Branch A evades nEDM but requires a second gauged flavon.) A null result at improved sensitivity disfavors branch B; a detection at the bound supports horn (i).
- **F4 — The lattice first-order verdict (S-1).** If a converged lattice run finds the transition is a **crossover** at any spacing, the out-of-equilibrium Sakharov leg is gone and the mechanism as constructed is falsified (§6.1).
- **F5 — Wall-velocity runaway (S-3).** If the full transport solve gives $v_w \to 1$ (runaway), the sourced asymmetry drops sharply, breaking the "no fine-tuning" claim and forcing a re-fit of m_3/H (§6.3).
- **F6 — Primordial tensors (horn selector).** Horn (i) predicts $r \leq 2\times10^{-11}$ ($r \approx 0$); *any* tensor detection kills horn (i) and selects the Candidate-C horns (ii/iii), which allow r. This is the tensor split of the two-horn theorem, cross-linked to the galactic face (a rising $a_0(z) \iff r \approx 0 \iff 1.78$ GeV DM).

X. RELATION TO THE PROGRAM

A. The single-source theorem and the two-horn structure

The dawn is a **two-horn theorem**, and the reason is a genuine, calculated EFT result:

> **Generalized single-source theorem.** In any clock-driven co-genesis, the chemical potential $\mu = H/2\pi$ cancels in the ratio Y_X/Y_B. Hence **η_B-calculable and m_X-sharp are mutually exclusive.**

Horn (i) (ECCG-res-iv) keeps $m_X = 1.78$ GeV sharp at the cost of η_B being a fit; horn (ii) (Candidate C) makes η_B calculable (because Σm_ν^2 is *measured*) at the cost of m_X becoming a band. The two horns are split observationally by primordial tensors (F6). Horn (iii) (glueball) *evades* the theorem because glueball DM is thermally sourced, not clock-sourced — the premise ($\mu = H/2\pi$ in both yields) fails — so it uniquely gets both a derived η_B and a predicted m_G, relocating the theorem's content from the DM mass to the DM abundance (the Br dial). The theorem's content is conserved, not evaded.

B. The clock does not source matter ($R_0 = 0$)

We restate the negative result of §I A because it is structurally important to the unification: the entropy clock $\mathcal{D}_E = (3/2)(1-w)$ links the *timing* and *thermodynamics* of the three faces, but it does **not** source the matter asymmetry. The candidate direct-source operator is $\mathbb{Z}_2$-even, so $R_0 = 0$ by symmetry (a calculated symmetry theorem). This is what makes ECCG a pre-inflationary initial-condition theory rather than a clock-pumped one, and it is why the program's honesty survives: the clock's unifying role is real (it is the same variable that reads $\mathcal{D}_E \to 3$ at dusk and sets $a_0 \propto (3/2)(1+w)$ galactically) without over-claiming a matter source it does not have.

C. Connection to dusk and the scale sector

Two threads connect this paper outward. **To dusk (SEDE):** ECCG's ~1.78 GeV ADM *is* SEDE's assumed cold dark matter, whose collapse drives the growth gate of the dark-energy model [Paper III / SEDE cosmology], and the identity $\omega_b \equiv \eta_B$ removes SEDE's baryon-density fit — so the union is more falsifiable than ΛCDM at equal continuous-parameter count. **To the scale sector:** horn (iii)'s glueball mass $m_G = 6\Lambda_h$ and (via the transmutation identity) the MOND scale a_0 are set by the same Λ_h that fixes the μ scale [Paper VI / scale sector], through the chain $\alpha_s(M_{\rm Pl}) \to \Lambda_h \to \mu \to a_0$. The dawn is thus the hinge of the program: it delivers the matter that dusk assumes and shares the confining scale that the galactic face reads.

XI. DISCUSSION AND LIMITATIONS

ECCG is, honestly, a **matched-EFT existence proof** of GeV-scale asymmetric co-genesis, not a fully predictive theory. Its strengths are real: one genuinely-calculated closure

($f_B = 28/79$), a self-consistent pre-inflationary history that survives reheating (D-1), a near-floor neutrino consistency constraint (Candidate C → NO, $\Sigma m_\nu \approx$ 59–64 meV), and a clean over-determination against the rest of the program. Its limitations are equally real and we do not hide them:

1. **The sign is environmental** — the magnitude of η_B is predicted, the sign is not (spontaneous CPV).
2. **The magnitude of η_B is a fit** — m_3/H is fitted, MaxEP fails by ~10^9, and the "four confirmations" share one calibrated normalization (§V D).
3. **The microscopic sector is benchmark/qualitative/preliminary** — first-order transition (lattice preliminary), SQCD vacuum (one-loop qualitative), v_w (LO ballistic). These are validated pipelines and specified projects, not confirmed results.
4. **The** Pin^+ **anomaly cancellation is constructed, not derived** (§VIII).
5. m_X**'s exact value depends on** $r_{X,BL}$ (1.78 GeV at $r_{X,BL} = 1$), and horns (ii)/(iii) trade the sharp mass for a band or a Br dial.

The correct summary is not "the dawn is calculated" but "the dawn is a coherent, over-determined, imminently falsifiable existence proof with one calculated closure, whose remaining ignorance is either irreducibly a fit (η_B magnitude, the sign) or purchasable with compute (S-1/S-2/S-3)."

XII. REPRODUCIBILITY

The results above reproduce from the accompanying analysis scripts in `https://github.com/spsingularity/eccg-matter-genesis` (a tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525535). Key scripts and reports:

- **Sphaleron / f_B:** `CLOSURES_assumed_vs_calculated.md` (N4); Harvey–Turner counting.
- **Spontaneous CP / Δ_{CP}:** `spontaneous_cp.py` → `SPONTANEOUS_CP_REPORT.md` (scan, `spontaneous_cp_scan.csv`, 40,000 draws).

- **Pre-inflation reheating cure (D-1):** `sims/preinflation_reheat_domain_check.py` → `D1_result_preinflation_reheating.md`.
- **Candidate C neutrino inversion (D-2):** `sims/candidateC_neutrino_inversion.py`, `candidateC_yield.py` → `D2_result_candidateC_neutrino.md`; figure `figures/candidateC_neutrino_inversion.png`.
- **Wall velocity (S-3):** `wall_velocity_precise.py` → `WALL_VELOCITY_PRECISE_REPORT.md`.
- **Global scan / benchmark point:** `GLOBAL_SCAN_REPORT.md`.
- **Flavor / S_3 gauging /** Pin$^+$ ν mod 16**:** `FLAVOR_MODEL_REPORT.md`, `S3_GAUGING_REPORT.md`, `PIN16_assignment_constructed.md`.
- **Glueball DM horn:** `sims/glueball_simp_relic.py` → `GLUEBALL_DM_assessment.md`.
- **External calculation specs (S-1/S-2/S-3):** `EXTERNAL_CALC_SPECS.md` (decision rules).
- **η_B provenance / P3 meta-pattern:** `predict_eta_B.py`; `AUDIT_round2_sweep.md`, `CLOSURES_assumed_vs_calculated.md`.
- **Pre-registration:** `PREREGISTRATION_falsifier_matrix.md` (Zenodo `10.5281/zenodo.21415326`).

Companion papers: [Paper III / SEDE cosmology, Zenodo 10.5281/zenodo.21651614], [Paper V / USC framework, Zenodo 10.5281/zenodo.21525529], [Paper VI / scale sector, Zenodo 10.5281/zenodo.21652167], [Paper VIII / APDM galactic, Zenodo 10.5281/zenodo.21652176].

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